CENTRE FOR DEVELOPMENT OF ADVANCED COMPUTING

C-DAC, Pune

Internship project review

PERIOD

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DURATION

3 months

STUDENT

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GUIDE

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APPLICATION

C-DAC Unified RAG

REPOSITORY

C:\CDAC\RAG

INTERNSHIP PROJECT

Development of an Web Crawler for Automated Knowledge Base Creation and RAG-Based Information Retrieval

Page-level provenance, hybrid BM25 and vector retrieval, and evidence-grounded answers over the crawled C-DAC corpus and its PDFs.

DONE	PARTIAL	PENDING
PDF path and live index	OCR on 31 sparse pages	Labelled retrieval score

RECORDED IN THIS REPOSITORY

Index rag_corpus_v1 on Elasticsearch 8.12.2 holds 12,054 documents.
60 crawled PDFs, 1,479 pages, 2,880 text chunks.
Five saved answers from Llama 3.2 through Ollama. None abstained.

Problem statement

ENGINEERING PROBLEM

Given crawled pages and PDFs, answer a natural-language question from those sources, name the supporting passage, and abstain when the evidence is too weak.

01 A general model

A language model can write fluently and still miss a fact that exists only in a PDF acquired after training.

02 Long documents

A full PDF does not fit a context window on every question. The retrievable unit has to be a passage.

03 Messy page text

pypdf text carries hyphenation, control characters, and running headers that should not become search terms.

04 Sparse pages

A scanned page yields almost no characters. The code detects that and can call Tesseract when the binary is present.

05 Two retrieval failures

BM25 misses a paraphrase. A vector search can miss a rare identifier. Their raw scores are not on one scale.

06 Unsupported answers

A fluent completion can still go beyond the hits. The ask path checks hit count, a minimum score, and role strings first.

07 No audit trail

An answer without a chunk id and a page cannot be checked. Those fields are stored on the text chunk and copied into the source list.

08 Mixed corpus

The live index holds HTML and crawled PDFs together. A PDF-only question has to set --sources pdf.

Objectives and scope

Status labels are the ones used in the internship report: DONE, PARTIAL, and PENDING.

Objective	Status	Evidence in this repository
Accept a PDF and reject a missing or empty file	DONE	PDFProcessor.process_pdf
Identify the file by SHA-256 of its bytes	DONE	generate_document_id
Extract and clean page text	DONE	1,479 stored pages; 0 failed pages
Detect sparse pages and run OCR	PARTIAL	31 pages flagged; Tesseract was not on PATH
Turn embedded images into chunks	PARTIAL	Code exists; image_chunks_generated summed to 0
Heading-aware chunks with overlap and merge	DONE	2,880 PDF chunks; cyber PDF had 3 merges
Page numbers on text chunks	DONE	Cyber PDF hit lists page 1
Page numbers on summary records	PARTIAL	summary_to_indexable leaves the page list empty
384-d embeddings and Elasticsearch	DONE	rag_corpus_v1, 12,054 documents
BM25, cosine kNN, min-max hybrid 0.3/0.7	DONE	Default mode; five saved answers are hybrid
Abstain and return a source list	DONE	Ask path; the five demos did not abstain
Score a labelled set with nDCG	PENDING	metrics.py has hit rate, recall, and MRR only

System architecture

DONE

ACQUISITION

1 Crawl › 2 HTML › 3 PDF download › 4 Extract › 5 Chunk

INDEX AND ANSWER

6 Embed › 7 Index › 8 BM25 › 9 HNSW › 10 Fusion › 11 Gate › 12 Answer

Crawl

Scrapy spider cdac starts at https://cdac.gov.in/, stays on that domain, and writes HTML sections plus a PDF manifest.

PDF path

ingest_uploaded_pdf, process-pdfs, and PDFRAGPipeline.ingest_pdf share pypdf extraction, cleaning, and structural chunking.

Answer path

search_corpus, then grounding_failure, then the LLM at temperature 0. RAGAnswer carries the source list.

WHAT THE LIVE INDEX ACTUALLY HOLDS

kb.source is all, so rag_corpus_v1 mixes HTML and crawled PDFs. Of 12,054 documents, 6,076 are crawled_pdf and 5,978 are html.

The PDF portion is 2,880 chunks + 2,880 extractive summaries + 316 composite summaries. Uploaded-PDF artifacts in the scan: 0.

Knowledge-graph rows are added at question time when --include-kg is set. The five saved answers each have 5 hybrid hits and 5 graph hits.

Technology stack

Versions were read from this project's environment and from the cluster at localhost:9200.

Technology	Version observed	Role in this repository
Python	3.12.10	Runtime; Docker image python:3.12-slim-bookworm
Scrapy	Project spider cdac	Bounded crawl of cdac.gov.in
pypdf	5.9.0	Page text and PDF metadata
PyMuPDF	2.1	Page render and embedded images; import name fitz
pytesseract / Tesseract	0.3.13 / binary not on PATH	OCR only when the binary is present
Pillow	13.0.0	Open rendered page images
sentence-transformers	3.4.1	Local all-MiniLM-L6-v2, 384 dimensions
Elasticsearch	server 8.12.2; client 8.19.3	BM25, kNN, and storage
Pydantic / PyYAML / dotenv	2.14.3 / 6.0.3 / 1.2.4	Settings, models, and masked secrets
pytest	8.4.2	Unit and contract tests
Ollama model in the demo	Llama 3.2	The five stored answers; mock remains the default

Configuration

DONE

File
config/settings.yaml, then environment variables by upper-case field name.

Secrets
LLM_API_KEY and ELASTICSEARCH_PASSWORD are SecretStr. get_masked_dict hides them.

Index
http://localhost:9200, index rag_corpus_v1.

Embeddings
provider local, sentence-transformers/all-MiniLM-L6-v2, dimension 384.

LLM default
provider mock. A live call needs LLM_ENDPOINT. The saved demo used Llama 3.2.

RETRIEVAL AND CHUNK VALUES

mode hybrid, top_k 5, pool 20, num_candidates 50
keyword_weight 0.3, semantic_weight 0.7, must sum to 1
fusion_strategy min_max; rrf_k 60 is unused until strategy is rrf
figure boost 4.0, image boost 2.0
grounding_min_score 0.35

HTML chunks: 1200 / overlap 150 / min 80
PDF chunks: 1200 / overlap 100 / min 300
Near-duplicate threshold 0.92
Crawl: cdac.gov.in, max 500 URLs, delay 1.5 s, 3 retries

Web crawl and PDF acquisition

DONE

1 Seed

python run.py crawl starts Scrapy spider cdac at https://cdac.gov.in/.

2 Boundary

allowed domain cdac.gov.in, URL cap 500, response cap 5 MB, download delay 1.5 seconds.

3 HTML

Sections are extracted and chunked. The page itself is not indexed as raw HTML.

4 PDFs

Discovered PDFs are downloaded to data/pdfs with a manifest, then process-pdfs builds artifacts.

5 Retries

A failed fetch is retried. The crawl ledger and success.log / error.log record the outcome.

6 Index

python run.py index embeds the selected records and bulk-writes rag_corpus_v1.

WHAT THE STORED PDF CORPUS IS

All 60 PDF artifacts have source_kind crawled_pdf. Uploaded-PDF artifacts in that scan: 0. The files live under data/pdfs, including 82f38954477f2c53.pdf, the cyber-forensics document used in the walkthrough.

A separate ingest --pdf command copies one local file into data/raw and marks it uploaded_pdf. That path was not the source of the 60-file corpus.

PDF extraction and cleaning

DONE

Bytes

The file is hashed with SHA-256 before pages are read. The same bytes always produce the same document_id.

Pages

pypdf PdfReader.pages[i].extract_text() supplies the raw string. Page numbers stay 1-based.

Cleaning

clean_extracted_text repairs hyphenation and whitespace. remove_repeated_boilerplate drops running headers.

Blocks

detect_block_type separates headings from paragraphs. The first section name, before any heading, is Introduction.

Failure

A page that cannot be read is stored as failed and the rest of the file continues. Across 60 PDFs, failed_pages is 0.

Record

The cyber PDF cc7477d8… opened 25 pages, 95 blocks, in 0.700 seconds of stored extraction time.

CORPUS EXTRACTION, SUMMED FROM THE 60 ARTIFACTS

1,479 pages. 1,448 counted successfully extracted. 0 failed. 31 still flagged requires_ocr. Every document’s extraction_status is SUCCESS, including pages left IMAGE_ONLY.

Sum of stored extraction durations: 44.38 seconds. Mean 0.74 seconds. Maximum 3.165 seconds. That sum is the artifact field, not a new stopwatch run.

OCR and embedded images

PARTIAL

WHAT THE CODE DOES

A page with very little extracted text sets requires_ocr.

If Tesseract and PyMuPDF are available, the page is rendered at 300 dpi and recognised.

Embedded rasters at least 50 px are saved under data/extracted/images.

OCR text of at least 15 characters can become an image chunk.

If the binary is missing, the processor logs a warning and leaves the sparse page unchanged.

WHAT THIS CORPUS SHOWS

31 pages remain flagged as requiring OCR.

The test re-run found the Tesseract binary not on PATH.

image_chunks_generated across the 60 PDFs is 0.

The Docker image installs tesseract-ocr. The host run that produced these artifacts did not use it.

No screenshot-answer from this corpus is claimed.

STATUS

OCR decision and fallback: PARTIAL. Image extraction to chunks: PARTIAL. Tests cover the graceful path when OCR cannot run. They do not certify recognition of the 31 sparse pages.

Context-aware chunking

DONE

STRUCTURAL RULE

```
section := Introduction
a heading closes the group and names the next section
emit when the next block would pass 1,200 characters
carry the last 100 characters, trimmed to a word
reject near-duplicates at similarity 0.92
merge same-section chunks under 300 characters
ceiling is 1.3 × 1,200 = 1,560 characters
```

WORKED PDF, NOT A SYNTHETIC EXAMPLE

File 82f38954477f2c53.pdf

25 pages, 95 blocks in.

61 chunks out. Mean length 842.

Minimum 89, maximum 1,339.

3 merges, 0 duplicates, 0 LLM calls.

Chunking time stored: 0.304 seconds.

The maximum above 1,200 is the merge ceiling.

ALL 60 PDFS

Text-chunk lengths: minimum 49, median 1,073, mean 933.6, maximum 1,481 characters. Sum of stored chunking durations: 15.73 seconds.

HTML uses a different setting: max 1,200, overlap 150, minimum 80, composite group size 10. LLM-assisted organisation exists in code and is off. The cyber artifact records llm_calls_made as 0.

Limitation recorded in the report: the heading detector can fire on a short line that is not a heading, and the overlap is a character tail.

Metadata, summaries, and lineage

Field	Meaning	Where it is set
document_id	SHA-256 of the PDF bytes	generate_document_id
chunk_id	doc_<id>_chunk_<nnn>	Chunker, renumbered after merge
page_numbers	Pages of the blocks in the group	Text chunks only
section	Heading that owned the group	Chunker
source_kind	html, crawled_pdf, or uploaded_pdf	Ingestion
content_kind	chunk, summary, or composite	Indexer
source_url, depth	Crawl lineage for downloaded files	PDF manifest and HTML extractor

SUMMARY HITS HAVE NO PAGE

Each PDF chunk also has an extractive summary. The demo summary that matched chunk 002 stored an empty page list, at the same fused score.

LINEAGE ON THE CYBER PDF HIT

pdf id 82f38954477f2c53, parent page the tenders EOI listing, depth 2, downloaded 11 November 2026.

Embedding generation

DONE

Chunk text › all-MiniLM-L6-v2 › 384 floats › dense_vector

One model both ways

Queries and documents use the active provider. The active provider is local. Mock and custom HTTP providers exist and are not selected.

Dimension check

The encoder’s width must equal embedding_dimension, which is 384. A mismatch is rejected before indexing.

Cosine

The code does not set normalize_embeddings. Cosine is declared on the Elasticsearch field, and the cluster normalises for that similarity.

What is not stored

The cyber walkthrough says the vector is 384 floats in the index. The vector itself is not printed in the report or the demo JSON.

NO PROJECT SIMILARITY SCORE IS INVENTED HERE

MiniLM is a compact English encoder. A long technical chunk becomes one 384-dimensional vector. Changing the model means a new index name, because an existing index dimension is not silently reused.

Elasticsearch knowledge base

DONE

12,054	6,076	5,978	5,616	5,616	822
Index documents	crawled_pdf	html	content chunks	summaries	composites

Mapping fact	Value in this project
Index name	rag_corpus_v1
Server	Elasticsearch 8.12.2, single node, Docker Compose
Text field	BM25 match query; the project does not reimplement BM25
embedding	dense_vector, dims 384, index true, similarity cosine
HNSW m and ef_construction	Not set. They remain Elasticsearch defaults
Document _id	The chunk identifier, so a re-index overwrites
Heap	512 MB JVM, from docker-compose.yml

BM25 keyword retrieval

DONE

$$BM25(D, Q) = \Sigma\ IDF(qi) \times [tf \times (k1+1)] / [tf + k1 \times (1 - b + b \times |D| / avgdl)]$$

The project sends an Elasticsearch match query. It does not set k1 or b.

qi

A query term after analysis.

tf

How often that term occurs in the chunk.

IDF

A term that appears in few chunks counts more.

|D| / avgdl

Length normalisation. Long chunks are pulled back.

WHERE IT IS USED

Exact names, programme titles, and identifiers. The hybrid call runs this list before fusion. A standalone BM25 ranking of the five demo questions was not saved.

WHERE IT MISSES

A paraphrase that shares almost no terms with the passage. That is why the default mode also runs the vector retriever.

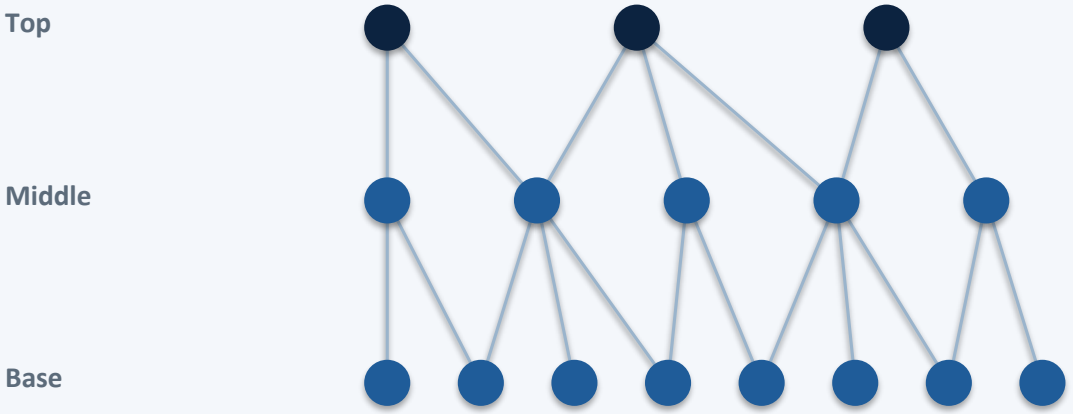
REFERENCE BOOST

When a question names a figure or table number, the BM25 query can add a terms clause, and fusion later adds figure_reference_boost 4.0. The five saved questions do not exercise that boost, so no boosted score is shown.

Vector retrieval and HNSW

DONE

SEARCH THE PROJECT ACTUALLY CALLS



Elasticsearch builds the graph. This code supplies the query vector, k, and num_candidates 50.

Query

The question is embedded with the same local model, then sent as a knn clause.

Similarity

Cosine, on the dense_vector field. A smaller angle ranks higher.

Not measured

m and ef_construction are unset. hnsw_benchmark.py can compare a NumPy scan with kNN. No output file is stored.

Scale

12,054 vectors of 384 floats are about 18.5 MB of raw payload before index overhead. That is a size calculation, not a disk reading.

STATUS

The index setting is DONE. A saved HNSW recall or latency number is NEEDS LIVE VALIDATION. The five demo JSON files do not store a separate vector ranking.

Hybrid retrieval

DONE

Pool 20 › Two scores › Min-max › 0.3 / 0.7 › Boosts › Top 5

Final = 0.3 × normalised BM25 + 0.7 × normalised vector + reference boost + image boost

MIN-MAX, THE ACTIVE STRATEGY

s' = (s − min) / (max − min), separately for each list.

If every score in a list is equal, each becomes 1.

A chunk missing from one list scores 0 on that side.

BM25 is unbounded and cosine is not, so the raw scores are not added.

RRF IS IMPLEMENTED AND INACTIVE

RRF = Σ 1 / (60 + rank), ranks starting at 1.

It runs only when fusion_strategy is rrf. The setting is min_max.

The same boosts are added after either strategy. Ties break on chunk_id.

ARITHMETIC ILLUSTRATION FROM THE REPORT, NOT A SAVED RETRIEVAL

BM25 scores 12, 6, and 9 normalise to 1, 0, and 0.5. Vector scores 0.40, 0.90, and 0.65 normalise to 0, 1, and 0.5. Hybrids are 0.30, 0.70, and 0.50. The second item ranks first because the semantic weight is larger.

The demo JSON stores only the fused score. It does not store the BM25 and vector components, so the live scores are not fitted to this example.

Grounding and answer generation

DONE PARTIAL

1 Use only the retrieved context.

2 Do not use outside knowledge.

3 Keep the source terminology.

4 If the context is insufficient, use the fixed refusal sentence.

5 Name the document and the page when the context supports the answer.

ASK-PATH GATE

No hits.

Leading score below 0.35.

A named role is absent from the hit text.

Temperature on a call that passes is 0.

DEFAULT CLIENT

settings set llm_provider to mock. That client returns a fixed sentence and is not evidence of answer quality.

STORED DEMO

responses.md names Llama 3.2 through Ollama. Five questions, hybrid, include_kg true, none abstained.

Worked query: cyber forensics

What cyber forensics work does C-DAC describe?

File
data/pdfs/82f38954477f2c53.pdf was already extracted to data/extracted/pdf_82f38954477f2c53.json.

Pages
25 pages, 0 failed, 0 OCR-flagged. 95 blocks became 61 chunks. Mean length 842. 3 merges. 0 LLM calls.

Hit
chunk_002, section “Expression of interest for reselling software and hardware products”, page 1.

Score
Hybrid fused score 0.6678465341224721. Its summary is a second hit at the same score, with no pages.

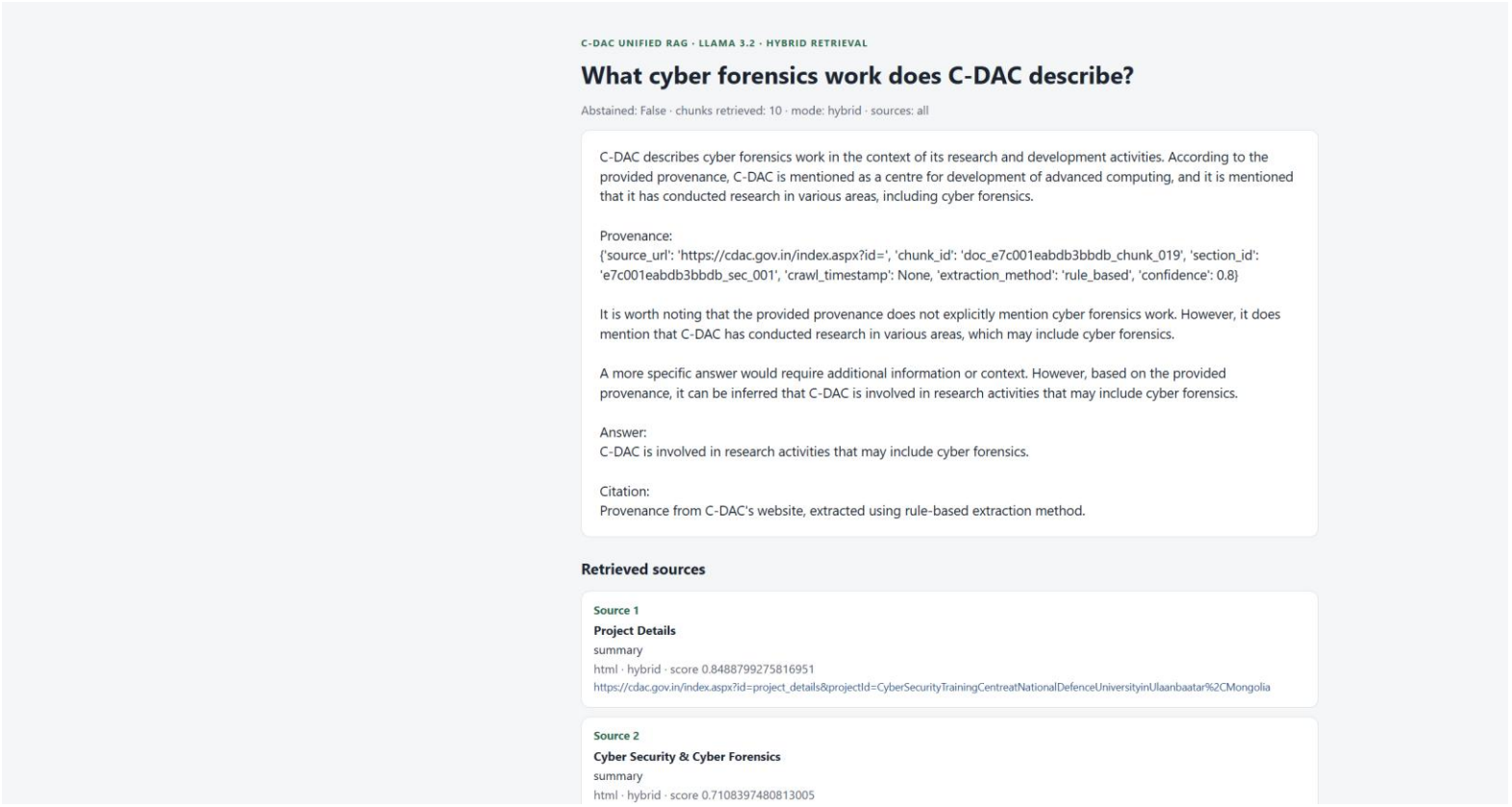
Gate
The leading score is above 0.35, so the model was called. Five knowledge-graph rows were then added.

Citation
The application source for the PDF is page 1. The model’s own provenance paragraph names a different chunk.

GENERATION GAP, RECORDED AND NOT REPAIRED

Llama 3.2 says the cited provenance does not explicitly mention the work, then still infers that C-DAC’s research may include cyber forensics. Retrieval found a real page. The extra inference is a generation-quality gap.

Saved answer page



Q5, FROM q5.json

Abstained: false

Hits: 10

Mode: hybrid

Model: Llama 3.2

Source 1

HTML summary, 0.8489

Project Details

Source 2

HTML summary, 0.7108

Cyber Security & Cyber Forensics

Source 3

Crawled PDF, page 1

Score 0.6678

Screenshot file: q5-cyber-forensics.png

Five recorded questions

#	Question	Abstain	Hits	Source mix
1	What is C-DAC's Aadhaar Data Vault service?	no	10	HTML only
2	Where is the C-DAC Pune centre located?	no	10	9 HTML, 1 PDF
3	What has C-DAC done in High Performance Computing?	no	10	HTML only
4	What training does the C-DAC ACTS programme provide?	no	10	9 HTML, 1 PDF
5	What cyber forensics work does C-DAC describe?	no	10	8 HTML, 2 PDF

Aadhaar, question 1

Top HTML chunk score 0.8853512993505596, section “Aadhaar Data Vault (ADV) as a service”.

Every question

Retrieval mode hybrid. Five corpus hits plus five knowledge-graph hits. include_kg was true.

Not a benchmark

BM25-only and vector-only lists for these questions were not saved. No precision or nDCG is stated.

Corpus results

Counted from data/extracted, the live index, and docs/demo-results. Query latency was not stored.

60 Crawled PDFs	**1,479** Pages	**0** Failed pages	**31** Pages needing OCR	**2,880** PDF text chunks
316 PDF composites	**12,054** Index documents	**384** Vector dimensions	**44.38 s** Extraction sum	**15.73 s** Chunking sum

HOW TO READ THESE NUMBERS

6,076 crawled-PDF index documents equal 2,880 chunks + 2,880 summaries + 316 composites. That is the check that the artifact scan and the index agree.

Content kinds across HTML and PDF: 5,616 chunks, 5,616 summaries, 822 composites.

These figures show that ingestion and indexing ran. They are not a retrieval-accuracy benchmark. Hit rate, recall, MRR, and latency have code and no stored scored run.

Repository and commands

DONE

Command	Result
python run.py status	JSON with elasticsearch_available; exit 0 even if the cluster is down
python run.py crawl	Scrapy spider cdac, domain and URL cap from config
python run.py process-pdfs	Artifacts for manifest PDFs, source_kind crawled_pdf
python run.py ingest --pdf FILE	One uploaded PDF artifact and chunk count
python run.py index	Embed and bulk-index; prints successfully_indexed
python run.py build-kg	Nodes and edges in data/kg/graph_v1.json
python run.py query "..." --sources pdf	Hits and the grounding flag, no LLM call
python run.py ask "..." --include-kg	RAGAnswer JSON; graph rows only when the flag is set
python run.py ui --port 8765	Inspector on 127.0.0.1
python -m pytest tests -q	Recorded re-run: 281 passed, 2 skipped, 7 teardown errors

Package rag has the crawl, extract, index, retrieve, answer, graph, eval, and UI modules. Elasticsearch is docker compose up -d elasticsearch.

Testing and validation

Re-run while the report was written: python -m pytest tests -q --tb=no. 281 passed, 2 skipped, 7 errors, 74.85 seconds.

Area	What a pass shows	What it does not show
PDF extraction	Multi-page text, empty pages, per-page failure	Teardown hit WinError 32 on a locked temp file
OCR	The pipeline continues when OCR cannot run	Tesseract was absent, so the 31 pages were not recognised
Chunking and references	Split, overlap, merge, figure and table parsing	No live cluster
Embeddings and index	Mock vectors, dimension checks, mapping shape	The stand-in client is not the live 12,054-document index
BM25, vector, hybrid	Query shape, fusion arithmetic, weight sum	No saved ranking of the five demo questions by mode
Ask contract	Prompt text, source filters, abstain without a model	No live Llama call inside the suite

EARLIER NOTE IN CURRENT_STATE.md

That file records 280 passed and 3 skipped. Chapter 27 uses the later re-run of 281 passed, 2 skipped, and 7 teardown errors. The test bodies passed; the errors are at path.unlink.

Limitations

OBSERVED

- 31 pages still require OCR. The test host had no Tesseract binary.
- No image chunk was produced for the 60 PDFs.
- pypdf does not recover column order.
- Heading and header rules are regular expressions and can misfire.
- Summary documents are retrieved without page numbers.
- One outlier in a pool of 20 rescales every other min-max score.
- The default LLM is a mock. Quality depends on the configured endpoint.
- Question 5’s model inferred past the hits after the gate had passed.
- Windows teardown can raise WinError 32 while PyMuPDF holds the file.

STRUCTURAL, NOT A MEASURED FAILURE RATE

- A 384-d MiniLM vector can miss a technical paraphrase or an exact identifier. Hybrid reduces that risk and does not remove it.
- Elasticsearch is required for a live question. status reports an outage and still exits 0.
- The index mixes HTML and PDF. A PDF-only question must pass --sources pdf.
- There is no measured limit for a PDF of many hundreds of pages.
- Hit rate, recall, MRR, and latency have no stored run on this corpus.
- Precision and nDCG are not implemented.
- HNSW recall against a brute-force scan was not saved.
- Query time and model time are absent from the demo JSON.
- Agent tool routing is not implemented. rag.services exposes ordinary functions.

Future scope

PENDING

Proposal	Gap it addresses	Phase
Run OCR on the 31 sparse pages	Tesseract was not available on the test host	1
Copy page numbers onto summaries	Summary hits cite no page	1
Close the PyMuPDF handle before unlink	WinError 32 during test teardown	1
Layout-aware extraction	Columns and tables are plain text	1
Cross-encoder over the pool of 20	Fusion has no second reader	2
Query rewriting	One string is embedded	2
Citation check on the completion	Question 5 inferred past the hits	2
Labelled set and hit rate, recall, MRR	metrics.py has no stored run	2
A documented --sources pdf preset	The demo searched all sources	3

Every row is proposed. None of it is described as already running. Agentic orchestration is outside the current code.

Conclusion

WHAT THIS INTERNSHIP BUILT

A crawler collects pages and PDFs from cdac.gov.in. Each PDF is hashed, extracted page by page, cleaned, and packed into chunks that keep their pages. Those chunks are embedded with all-MiniLM-L6-v2 and stored in Elasticsearch beside a BM25 field and a cosine vector.

Questions use a min-max hybrid of BM25 and vector search, weights 0.3 and 0.7, a pool of 20, and a final cut of 5. The ask path can abstain. It returns the source list that retrieval actually produced.

The stored corpus is 60 PDFs, 1,479 pages, and 2,880 chunks, inside an index of 12,054 documents. Five Llama 3.2 answers are the generation record. 281 tests passed, with seven Windows teardown errors that did not fail the assertions.

STILL OPEN

OCR on the sparse pages, page numbers on summary hits, a labelled retrieval score, and a check that the model’s sentences stay inside the source list.

THANK YOU — QUESTIONS AND DISCUSSION

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